Cyber Warfare & Advanced Persistent Threats

Homework Assignment — Case Study: Stuxnet

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Date Assigned:	23-07-2026	**Date Due:**	

Instructions

Read the assigned materials on cyber warfare and Advanced Persistent Threat (APT) groups, then complete Parts A, B, and C below. Answer in complete sentences. Total marks: 50.

Part A — Understanding Cyber Warfare (10 marks)

1. **In your own words, define "cyber warfare." How is it different from ordinary cybercrime or hacktivism? (3 marks)**

   The use of cyber attacks is a cyber wwarfareby a nation-state or state-sponsored group to disrupt, damage, or spy on another country's systems or critical infrastructure. Unlike ordinary cybercrime, which is mainly motivated by financial gain, and hacktivism, which is driven by political or social protest, cyber warfare focuses on
   achieving military, political, or strategic national objectives.

2. **List and briefly explain THREE characteristics that make cyber warfare attractive to nation-states compared to conventional warfare. (3 marks)**

• Low cost compared to conventional warfare.
• Difficult attribution, allowing attackers to deny responsibility.
• Ability to target critical infrastructure remotely with minimal risk to personnel.

3. **Name and briefly describe TWO real-world examples of cyber warfare incidents (other than Stuxnet). (4 marks)**

1. 2015 Ukraine Power Grid Attack – Attackers disrupted electricity for thousands of people.
2. NotPetya (2017) – Malware spread globally, causing billions of dollars in damage while initially targeting Ukraine.

Part B — Advanced Persistent Threats (APTs) (15 marks)

4. **What is an Advanced Persistent Threat (APT)? What makes an APT "advanced" and "persistent"? (3 marks)**

   An APT is a long-term, targeted cyberattack conducted by skilled attackers. It is 'advanced' because it uses sophisticated techniques and exploits, and 'persistent' because attackers maintain long-term access while avoiding detection.

5. **Choose THREE APT groups from your reading materials. For each, identify: (a) its believed country of origin, (b) one notable attack or campaign it is linked to. (9 marks)**

   → **APT28 (Fancy Bear)**
   - **Country of Origin:** Russia
   - **Notable Attack:** 2016 U.S. DNC cyberattack.

   → **APT1 (Comment Crew)**
   - **Country of Origin:** China
   - **Notable Attack:** Cyber espionage campaigns targeting global companies to steal intellectual property.

   → **Lazarus Group**
   - **Country of Origin:** North Korea
   - **Notable Attack:** 2014 Sony Pictures Entertainment cyberattack.

6. **Why is it often difficult to formally prove which country or group is behind an APT attack? Give two reasons. (3 marks)**

   Attackers hide their identities using proxies, VPNs, botnets, and compromised systems. They can also reuse tools from other groups to create false attribution, making conclusive proof difficult.

Part C — Case Study: Stuxnet (25 marks)

Using your case study notes on Stuxnet, answer the following questions.

7. **What was the target of the Stuxnet attack, and what was the attacker's strategic objective? (3 marks)**

   Stuxnet targeted the Natanz uranium enrichment facility in Iran. Its objective was to slow Iran's nuclear program by secretly damaging uranium-enrichment centrifuges.

8. **Describe, step by step, how Stuxnet is believed to have entered the Natanz facility despite it being air-gapped** (not connected to the internet). (4 marks)

   It is believed Stuxnet entered through infected USB flash drives brought into the facility. After execution, it spread across Windows systems until it reached Siemens industrial control systems.

9. **Explain how Stuxnet was able to damage the centrifuges while operators saw no signs of a problem on their monitoring systems.** (4 marks)

   Stuxnet altered PLC commands to repeatedly change centrifuge speeds, causing physical damage. At the same time, it sent normal sensor readings to monitoring systems so operators believed everything was functioning normally.

10. **What is a "zero-day vulnerability"? Why was it significant that Stuxnet used four of them in a single operation?** (4 marks)

A zero-day vulnerability is a software flaw unknown to the vendor and without a patch. Using four zero-days made Stuxnet highly effective, difficult to detect, and capable of spreading before defenses were available.

11. **Describe TWO measurable impacts of the Stuxnet attack on Iran's nuclear program. (4 marks)**

• Around 1,000 centrifuges were damaged or destroyed.
• Iran's uranium enrichment program was delayed significantly, requiring equipment replacement and recovery.

12. **In your opinion, should a cyberattack that causes physical destruction (like Stuxnet) be treated as an "act of war" under international law? Justify your answer in a short paragraph (100–150 words).** (6 marks)

In my opinion, a cyberattack like **Stuxnet** should be treated as an act of war if it causes physical destruction or seriously threatens a country's critical infrastructure. Unlike ordinary cyberattacks that steal data or disrupt services temporarily, Stuxnet physically damaged industrial equipment and affected a nation's strategic nuclear program without using conventional weapons. I believe such attacks can have consequences similar to military strikes by impacting national security, public safety, and the economy. However, I also think governments should not rush to label every major cyberattack as an act of war. They must first examine the extent of the damage, the attacker's intent, and reliable evidence, since identifying the real attacker can be difficult. International cooperation and clear legal rules are essential to ensure fair and proportionate responses.

Grading Rubric

Section	Marks Available	Marks Awarded	Comments
Part A — Cyber Warfare	10		
Part B — APT Groups	15		
Part C — Stuxnet Case Study	25		
Total	**50**		

Submission note: Please submit a typed or clearly handwritten copy by the due date shown above.